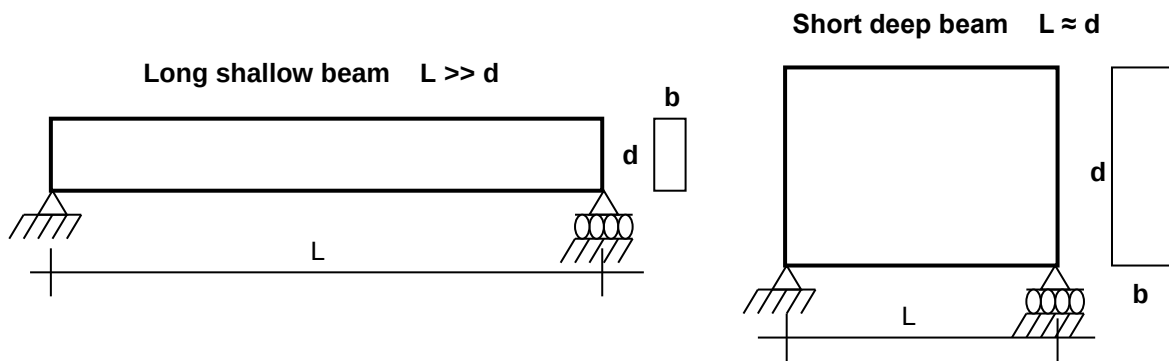


FCC3 232 CHAP 2 SHEAR STRESS IN BEAMS

2.1 INTRODUCTION

Beams can be divided into two categories namely **long shallow beams** and **short deep beams**. Long shallow beams have a **span which is much larger than the section depth**, while short deep beams have a **span which is comparable to the section depth** as illustrated in the diagrams below. Short deep beams are outside the scope of FCC3 232



Long shallow beams are designed to carry predominantly **transverse loads**. The effects of the applied loads are **bending moments** and **transverse shear forces**.

When a beam section is subjected to a **transverse shear force**, vertical shear stress will be induced over that section. Horizontal shear stress will be induced over the section in order to compliment the vertical shear stress. The **vertical** and **horizontal complimentary shear stresses are equal in magnitude**. The magnitude of the shear stresses will depend on the **size and shape of the section** and the magnitude of the **applied transverse shear force** at that section.

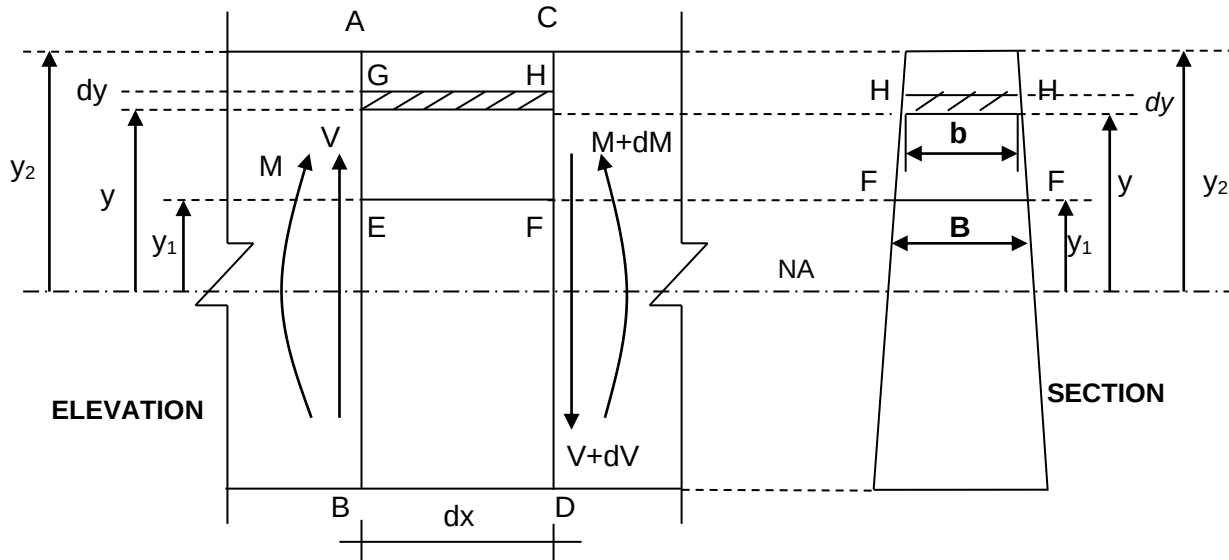
In this chapter, we shall derive **Dmitrii shear stress** equation which is used to determine the **shear stress** induced in a beam section due to an applied transverse shear force. We shall then **use the equation** to determine the **shear stress distribution over some commonly used sections**.

The following **assumptions** are made in the **theory of shear stresses**.

1. The beam section is **prismatic**, the material is **homogeneous, isotropic, linearly elastic**, and is not stressed beyond the **elastic limit**.
2. The **shear stress is uniform** across the **width of the section** at any level from the neutral axis AND the **shear stresses do not affect the flexural stresses**.

2.2 DMITRII SHEAR STRESS EQUATION

Let AB and CD be adjacent cross sections of a beam distance dx apart. The forces acting on AB are V and M , while the forces acting on CD are $V+dV$ and $M+dM$ as shown in the diagram below. We shall develop an equation for the shear stress at the level EF.



Bending stress at distance y from NA

$$f = My/I$$

Horizontal force at end G of element GH

$$F_G = f dA = (My/I) b dy$$

Horizontal force on end H of element GH

$$F_H = f dA = (M+dM) (y/I) b dy$$

Net horizontal force on end H of element GH

$$F_H - F_G = (dM/I) y b dy = (dM/I) y dA$$

[In the above equations, $dA = b dy$ is the elemental area of the cross section at level H].

Total horizontal force at level EF is obtained by integration from level EF to level AC

$$\text{Force} = (dM/I) \int_{y_1}^{y_2} y dA$$

If we define **a = area of the cross section above level EF**

And **$\tilde{y}$ = distance of the centroid of this area from the NA**

Then $\int_{y_1}^{y_2} y dA = a \tilde{y}$ (the first moment of the area) and $\text{Force} = (dM/I) a \tilde{y}$

The shear stress at level EF = Force/Area = $(dM/I) a \tilde{y} / (B dx) = (1/IB)(dM/dx) a \tilde{y}$

In theory of structures, $dM/dx = V$, the shear force.

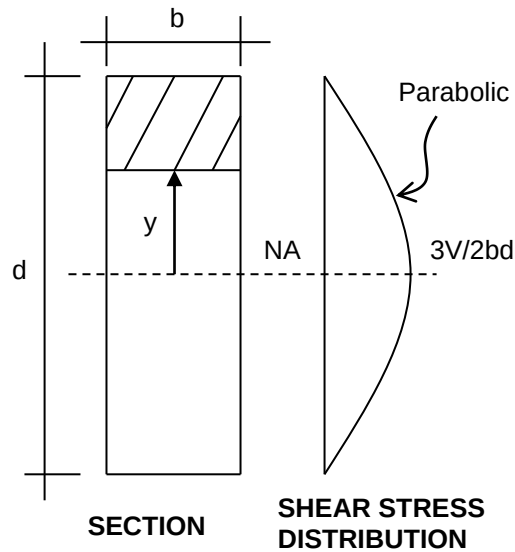
The **shear stress at level EF, denoted by τ** , is therefore given by:

$$\tau = (V/IB) a \tilde{y} \quad \text{-----} \quad \text{Eqn 1}$$

Eqn 1 is called the **Dmitrii shear stress equation** and was first derived by **Dmitrii Ivanovich Zhuravskii** in **1855**.

2.3 CASE STUDY 1 RECTANGULAR CROSS SECTION

Consider the solid rectangular section shown below



At any distance y from NA

$$a = b(d/2 - y) \quad \text{and} \quad \tilde{y} = (1/2)(d/2 + y)$$

$$a \tilde{y} = (b/2)(d/2 - y)(d/2 + y) = (b/2)(d^2/4 - y^2)$$

Using **Dmitrii shear stress equation** the shear stress is given by

$$\tau = (V/IB) (b/2)(d^2/4 - y^2)$$

For the rectangular section, $I = bd^3/12$ and $B = b$

The shear stress at level y is therefore given by

$$\tau = (12V/b^2d^3) (b/2)(d^2/4 - y^2)$$

$$\tau = (6V/bd^3)(d^2/4 - y^2) \quad \text{----- Eqn 2}$$

Eqn 2 shows a parabolic distribution of the shear stress with respect to the distance y .

At the **top and bottom** fibers $y = d/2$ and the shear stress $\tau = 0$

At the **NA**, $y = 0$ and the shear stress attains a maximum given by $\tau_{\max} = 3V/2bd$

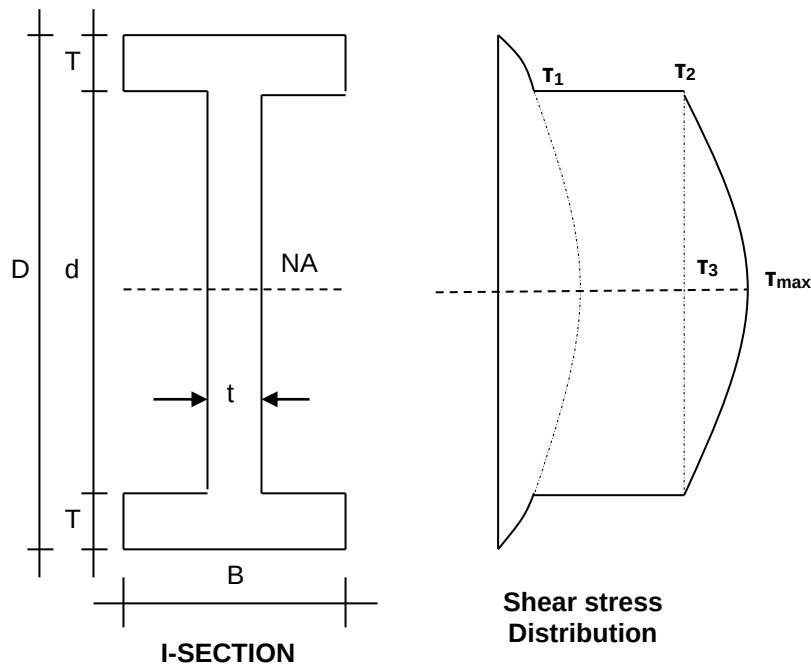
If the **mean shear stress** over the section is calculated as $\tau_{\text{mean}} = V/bd$, then the **maximum shear stress** is one and half times the mean shear stress.

$$\tau_{\max} = (3/2) \tau_{\text{mean}}$$

NOTE: The shear stresses are considered **positive** if the **shear force V is positive**.

2.4 CASE STUDY 2 I-SECTION

Consider the I-section shown below. Due to the geometry of the section, two scenarios must be considered. **Scenario 1: y within the web** and **scenario 2: y within the flange**



2.4.1 Scenario 1: y within the web

The quantity $a\tilde{y}$ is calculated in two parts

$$\begin{aligned} a\tilde{y} &= a_1 y_1 + a_2 y_2 \\ a\tilde{y} &= t(d/2 - y)(1/2)(d/2 + y) \\ &\quad + B(D/2 - d/2)(1/2)(D/2 + d/2) \\ &= (t/2)(d^2/4 - y^2) + (B/8)(D^2 - d^2) \end{aligned}$$

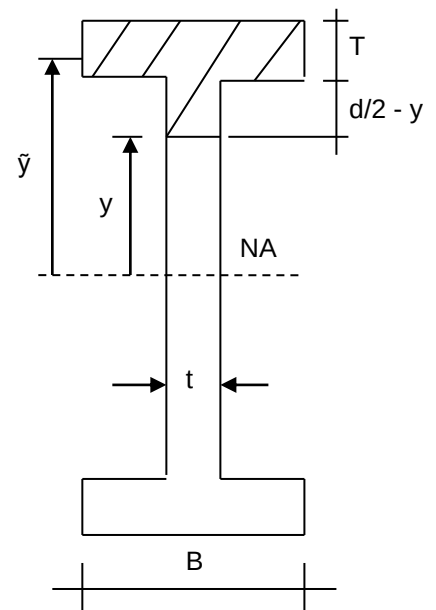
The value of I can be calculated as

$$I = (1/12)[BD^3 - (B - t)d^3], \text{ and the value of } B = t$$

Using **Dmitrii shear stress equation**

the shear stress is given by

$$\tau = (V/I t) [(t/2)(d^2/4 - y^2) + (B/8)(D^2 - d^2)] \text{ ----- Eqn 3}$$



Eqn 3 shows a parabolic distribution of the shear stress with respect to the distance y.

The maximum shear stress is induced at the NA when y = 0.

2.4.2 Scenario 2: y within the flange

$$a = B(D/2 - y) \quad \text{and} \quad \tilde{y} = (1/2)(D/2 + y)$$

$$\text{so that} \quad a\tilde{y} = (B/2)(D^2/4 - y^2)$$

The value of I can be calculated as

$$I = (1/12)[BD^3 - (B - t)d^3], \text{ and the value of } B = B$$

Using **Dmitrii shear stress equation**

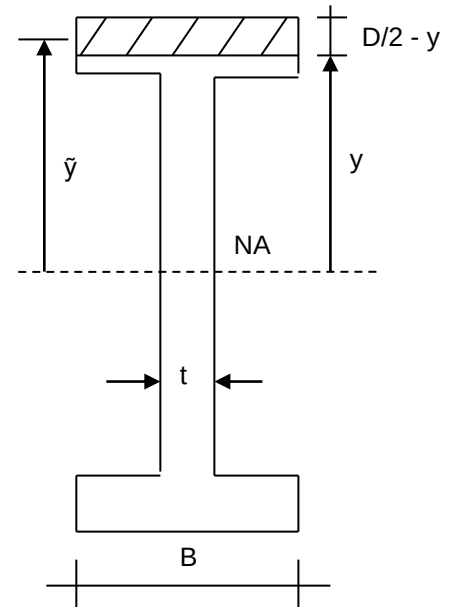
the shear stress is given by

$$\begin{aligned} \tau &= (V/I B) [(B/2)(D^2/4 - y^2)] \\ &= (V/2I)[D^2/4 - y^2] \quad \text{----- Eqn 4} \end{aligned}$$

Eqn 4 shows a parabolic distribution of the shear stress with respect to the distance y .

NOTE: The shear stress distribution will have a first order discontinuity (called a jump) at the junction between the web and the flange.

The overall shear stress distribution is plotted alongside the given I-section.



2.4.3 SHORT CUT

Let τ_1 = shear stress at the flange-web junction evaluated on the **flange side**

$$\tau_1 = (V/IB) a \tilde{y} = (V/IB) B T \frac{1}{2}(d + T) \quad \text{----- (1)}$$

Let τ_2 = shear stress at the flange-web junction evaluated on the **web side**

$$\tau_2 = (V/IB) a \tilde{y} = (V/It) B T \frac{1}{2}(d + T) \quad \text{----- (2)}$$

The **ratio of τ_2 to τ_1** is obtained by division $\tau_2/\tau_1 = B/t$

$$\text{Therefore} \quad \tau_2 = \tau_1 \times (B/t) \quad \text{----- (3)}$$

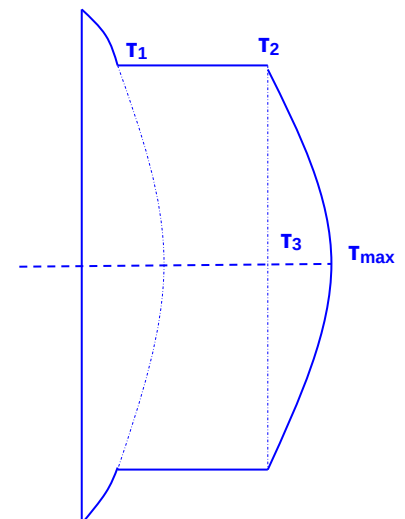
Let τ_3 = max shear stress induced in the web, if the web was considered alone

$$\tau_3 = (V/IB) a \tilde{y} = (V/It) (t d/2) (d/4) \quad \text{----- (4)}$$

[The I value is for the whole section]

The maximum shear stress induced in the web at NA is given by

$$\tau_{\max} = \tau_2 + \tau_3 \quad \text{----- (5)}$$



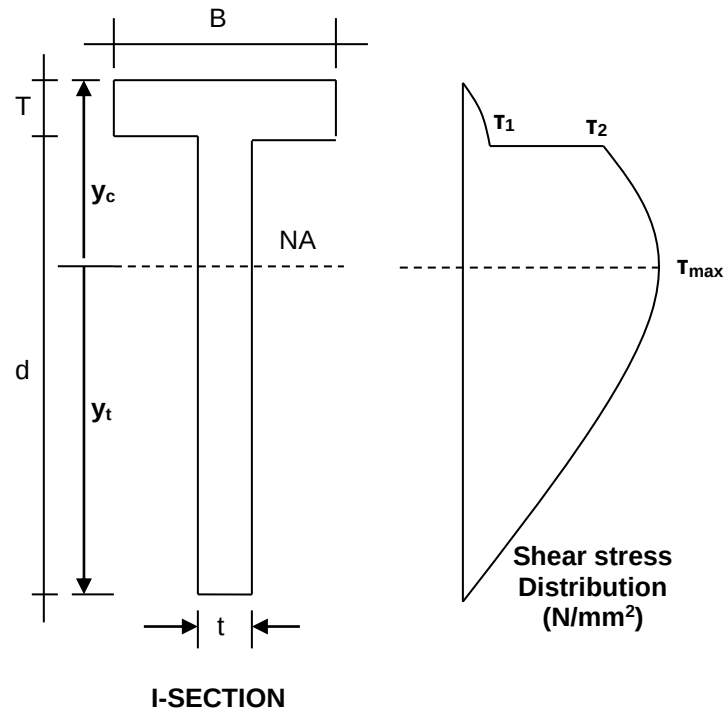
Shear stress Distribution

Conclusion: If you need to determine $\tau_{\max}$, then you need to: -

- 1) Evaluate τ_1 using Eqn (1) Obtain τ_2 using Eqn (3)
- 2) Evaluate τ_3 using Eqn (4) Determine $\tau_{\max}$ using Eqn (5)

2.5 CASE STUDY 3 T-SECTION

Consider the T-section shown below. Due to the geometry of the section, two scenarios must be considered. **Scenario 1: y within the web** and **scenario 2: y within the flange**.



2.5.1 Scenario 1: y within the web

[y can be measured from the bottom, for convenience]

The values of I , y_t and y_c are calculated in advance.

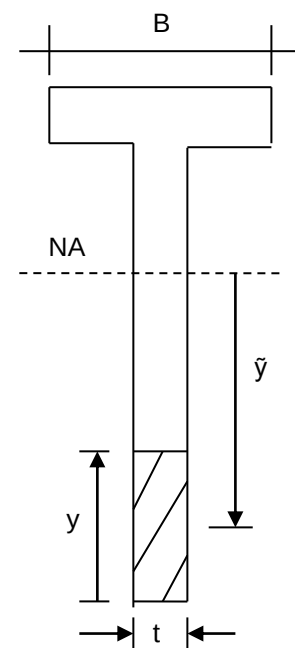
$$a = t y \quad \tilde{y} = y_t - y/2 \quad \text{and} \quad B = t$$

Using **Dmitrii equation** the shear stress is given by

$$\begin{aligned} \tau &= (V/I t) [t y (y_t - y/2)] \\ &= (V/I) y(y_t - y/2) \quad \text{----- Eqn 5} \end{aligned}$$

Eqn 5 shows a parabolic distribution of the shear stress with respect to the distance y .

Maximum shear stress occurs at NA when $y = y_t$.



2.5.2 Scenario 2: y within the flange

[y can be measured from the top, for convenience]

The values of I , y_t and y_c are calculated in advance.

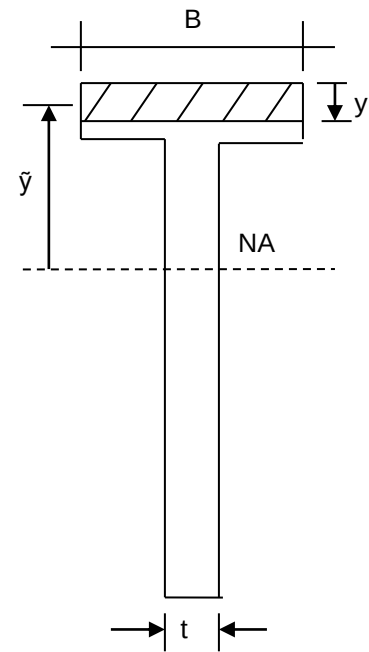
$$a = B y \quad \tilde{y} = y_c - y/2 \quad \text{and} \quad B = B$$

Using **Dmitrii equation** the shear stress is given by

$$\begin{aligned} \tau &= (V/I B) [B y (y_c - y/2)] \\ &= (V/I) y(y_c - y/2) \end{aligned} \quad \text{----- Eqn 6}$$

Eqn 6 shows a parabolic distribution of the shear stress with respect to the distance y .

[The shear stress distribution will have a first order discontinuity (called a jump) at the junction between the web and the flange]



The overall shear stress distribution is plotted alongside the given T-section.

2.6 WORKED EXAMPLES

Example 1: The timber section shown below is subjected to a shear force of 40 kN. Plot the shear stress distribution over the section indicating the maximum shear stress.

Solution $[V = 40 \times 10^3 \text{ N}]$

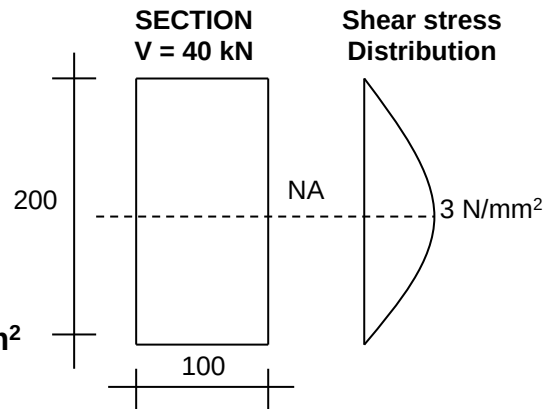
Using Eqn 2 the shear stress is given by

$$\tau = (6V/bd^3)(d^2/4 - y^2)$$

$$= 300 \times 10^{-6}(100^2 - y^2)$$

Maximum shear stress is induced at NA

when $y = 0$ $\tau_{\max} = 300 \times 10^{-6}(100^2) = 3 \text{ N/mm}^2$



Example 2: The steel I-section shown below is subjected to a shear force of 150 kN.

- Plot the shear force distribution over the section, showing all salient values.
- Calculate the ratio of maximum to mean shear stress.
- Calculate the percentage of shear force carried by the web.

Solution (a) Using Eqn 3 for y within web

$$\tau = (V/I) \left[\frac{t}{2} \left(\frac{d^2}{4} - y^2 \right) + \frac{B}{8} (D^2 - d^2) \right]$$

$$I = BD^3/12 - (B - t)d^3/12 = 227.57 \times 10^6$$

$$\tau = (65.9 \times 10^{-6}) [5(22.5 \times 10^3 - y^2) + 640 \times 10^3]$$

At NA, $y = 0$ and $\tau_{\max} = 49.6 \text{ N/mm}^2$

At the flange, $y = 150$ and $\tau = 42.2 \text{ N/mm}^2$

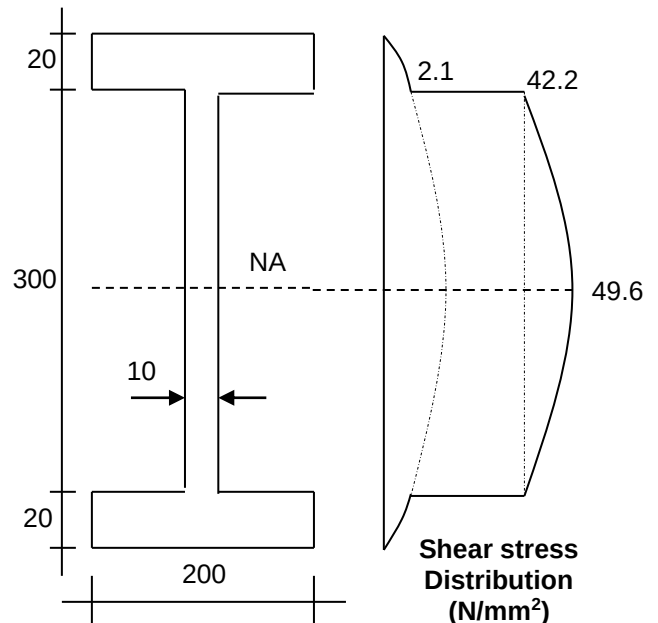
Using Eqn 4 for y within flange

$$\tau = (V/2I) [D^2/4 - y^2] \quad (I = 227.57 \times 10^6)$$

$$\tau = (329.6 \times 10^{-6}) [28.9 \times 10^3 - y^2]$$

At the top, $y = 170$ and $\tau = 0$

At the web, $y = 150$ and $\tau = 2.1 \text{ N/mm}^2$



Solution (b) $A = 200(340) - 190(300) = 11 \times 10^3 \text{ mm}^2$ $\tau_{\text{mean}} = V/A = 13.6 \text{ N/mm}^2$

$$\tau_{\max}/\tau_{\text{mean}} = 49.6/13.6 = 3.65$$

Solution (c) Shear force carried by web = Shear stress area over web $\times$ web width

$$= [42.2 + (2/3)(49.6 - 42.2)] \times 300 \times 10 = 141.4 \text{ kN}$$

$$\% \text{ shear force carried by web} = 141.4/150 \times 100 = 94.3\%$$

Example 3: A T-Beam section is fabricated by welding a web plate onto a flange plate as shown below. The section is subjected to a shear force of 47.7 kN.

- Plot the shear force distribution over the section, showing all salient values.
- Calculate the percentage of shear force carried by the web.
- Calculate the shear force transmitted per meter run in the welded connection

Solution (a)

$$A = 100 \times 15 + 150 \times 10 = 3 \times 10^3 \text{ mm}^2$$

$$y_c = [100 \times 15 \times 7.5 + 150 \times 10 \times 90] / A$$

$$= 48.75 \text{ mm}$$

$$y_t = 165 - y_c = 116.25 \text{ mm}$$

$$I = 100 \times 15^3 / 12 + 100 \times 15 (y_c - 7.5)^2$$

$$+ 10 \times 150^3 / 12 + 10 \times 150 (y_t - 75)^2$$

$$= 7.95 \times 10^6 \text{ mm}^4$$

Using Eqn 5 for **y within the web**

$$\tau = (V/I) y(y_t - y/2) = (6 \times 10^{-3}) y(116.25 - y/2)$$

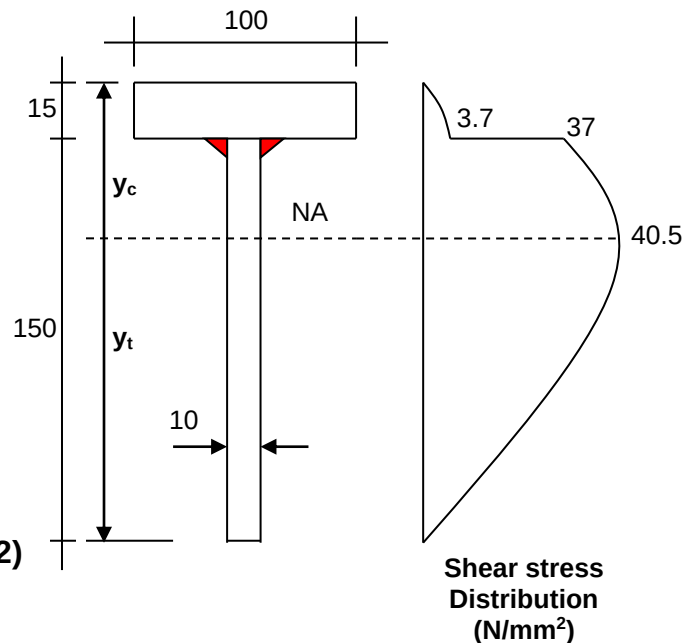
At the bottom fibers $y = 0$ and $\tau = 0$

At NA, $y = y_t$ and $\tau_{\max} = 40.5 \text{ N/mm}^2$

At the flange, $y = 150$ and $\tau = 37 \text{ N/mm}^2$

Using Eqn 6 for **y within flange** $\tau = (V/I) y(y_c - y/2) = (6 \times 10^{-3}) y(48.75 - y/2)$

At the top, $y = 0$ and $\tau = 0$ At the web, $y = 15$ and $\tau = 3.7 \text{ N/mm}^2$



Solution (b) Shear force carried by web = Shear stress area over web x web width

$$= 10 \times \int_0^{150} (6 \times 10^{-3}) y(116.25 - y/2) dy = 44.7 \text{ kN}$$

$$\% \text{ shear force carried by web} = 44.7 / 47.7 \times 100 = 93.7\%$$

Solution (c) Shear stress at the top of the web = 37 N/mm²

Shear force per meter = stress in web x width of web x 1 meter

$$= 37 \times 10 \times 1000 = 370 \text{ kN}$$